Yale University

Introductory Machine Learning

S&DS 265 / 565 · Course Information · Fall 2026

Zhuoran Yang

Fall 2026

AI usage: figures were generated, and these slides polished, with the help of AI tools.

Outline

1. Course and Staff
2. Prerequisites
3. Materials
4. Coursework and Grading
5. Collaboration and AI Policy

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The Course

S&DS 265 / 565

An introduction to machine learning for students across Yale, from linear regression through diffusion and language models. The emphasis is on why methods work, and every method is paired with a runnable demonstration.

lectures	Monday and Wednesday, 1:05–2:30 pm
location	Davies Auditorium, 15 Prospect Street
course site	Canvas
discussion	Ed
submission	Gradescope

Instructor

Zhuoran Yang

Assistant Professor, Statistics and Data Science

- Email: zhuoran.yang@yale.edu
- Office hours: Wednesdays 6–7 pm
- Kline Tower 1021

Research

reinforcement learning
deep learning
large language models
optimization
game theory

Teaching Fellows and Learning Assistants

role	name	email	office hours
TF	Zhiling Zhou	zhiling.zhou@yale.edu	Tuesday 6–7 pm
TF	Levon Papikyan	levon.papikyan@yale.edu	Wednesday 4–5 pm
ULA	Lucas Papamitsakis	lucas.papamitsakis@yale.edu	Monday 11:30 am–12:30 pm
ULA	Meka Rucker	meka.rucker@yale.edu	Tuesday 10:30–11:30 am
ULA	Ethan Hao	ethan.hao@yale.edu	Wednesday 11:30 am–12:30 pm

Office hours are the fastest way to get help. Come for help on a specific homework question, a specific error, or a specific confusion. Locations are posted on Canvas.

This team is subject to change.

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Prerequisites

Probability

distributions, densities, marginal and conditional probability

Statistics

mean, variance, maximum likelihood estimation

Linear algebra

vector and matrix calculus, inversion, eigenvalue and singular value decompositions

Algorithms and programming

Python, data structures, computational complexity

Lecture 2 reviews what the course uses. Linear algebra and multivariate calculus are required — students without them often struggle.

If You Are Unsure Whether to Take This Course

- The course is designed as an accessible but rigorous introduction for a wide range of students with SDS/CS/Eng or similar background
- Programming is central: assignments use Python in Jupyter notebooks
- We will also use math to rigorously build the ideas
- S&DS 365 / 665, Intermediate Machine Learning is the more technical follow-up

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Textbooks

No textbook is required. Three are recommended, and all three are free online.

- **ISLP** — James, Witten, Hastie, Tibshirani, Taylor, *An Introduction to Statistical Learning: with Applications in Python*.
  The classical core: regression, classification, resampling, trees. `statlearning.com`
- **D2L** — Zhang, Lipton, Li, Smola, *Dive into Deep Learning*.
  The computational path: data, model, loss, optimizer, in runnable code. `d2l.ai`
- **DL** — Goodfellow, Bengio, Courville, *Deep Learning*.
  Background and depth for the neural-network half. `deeplearningbook.org`

Downloaded to Canvas/Files/textbooks

The Slides Are the Primary Text

- The lecture slides are self-contained: every definition, derivation, and worked example you are responsible for is in them;
- The textbooks are for a second explanation, extra examples, and depth beyond what one lecture can hold;
- Each lecture ships with a **demo notebook** that reproduces every number and figure on the slides;
- Slides are posted before each lecture.

Software

- **Python** with NumPy, pandas, Matplotlib, scikit-learn, and PyTorch;
- **Google Colab** is recommended — nothing to install, and a free GPU when a lecture needs one;
- A local installation is fine; the course provides an environment file;
- Assignments are submitted as executed notebooks through Gradescope.

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What the Work Is

- **Five homework assignments**, each with a theory part — derivations and calculations by hand — and a coding part.
- **Two examinations**, both closed book and closed computer.
- **S&DS 565** students also complete extra problems on each assignment.

component	weight
five homework assignments	50%
midterm examination	25%
final examination	25%

Homework

- Five in total, roughly every two weeks;
- Released Wednesday, due Wednesday at 11:59 pm — both parts released together, so you can plan the whole window;
- Submitted through Gradescope, sometimes in separate portals for the coding and written parts;
- The theory part may be typed or neatly handwritten and scanned;
- The coding part must run start to finish in a fresh kernel.

Discussing approaches is encouraged. The write-up, the words, and the code must be your own.

Late Work

Wednesday at 11:59 pm is a real deadline. Late work is accepted for 72 hours, at 10% per 24 hours or part of one.

submitted	credit
by Wednesday 11:59 pm	100%
within 24 hours	90%
within 48 hours	80%
within 72 hours	70%
after that	not accepted

Ten minutes late costs 10%. Submit what you have. Illness or another serious circumstance needs a Dean's Excuse, which carries no penalty.

The Two Examinations

Midterm

- Wednesday 14 October tentative
- In class
- Covers Lectures 1–12

Final

- Tuesday 22 December
- Set by the Registrar
- Cumulative

Both are closed book and closed computer. Neither can be rescheduled, and no alternate sittings are offered.

The final is the last day of the examination period.

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Collaboration

Collaboration is allowed and encouraged, within limits.

- You may discuss a problem with at most one other person in the course;
- You may also discuss it with AI tools;
- You must acknowledge both — name the person, and name the tool and what you used it for;
- You must write your own solution: no shared text, no shared code;
- You must be able to explain every line you submit.

Discussing an idea is collaboration. Copying a solution is a violation.

Declaration

End every submission with one short block declaring both your collaborators and your use of AI tools:

Example

"I discussed Problems 2 and 3 with **[name]**. I used **[tool]** to debug the plotting code in Problem 4 and to check my derivation in Problem 1. I wrote all submitted work myself."

- Declaration is a professional norm; it costs nothing and is never held against you;
- Omitting it is itself the violation.

AI Tools

Use them for

- Syntax and plotting calls
- Explaining an error message
- Explaining something from lecture a second way (highly recommend)

Do not use them to

- Produce a solution to an assigned problem
- Submit work you cannot explain
- In either examination, where all devices are forbidden

Please turn off AI autocomplete while working on assignments. Ordinary completion of a function name is entirely fine.

Academic Integrity and Support

- Yale's academic integrity policy applies in full; when in doubt, ask before you submit;
- Accommodations: register with Student Accessibility Services — sas.yale.edu, sas@yale.edu, 203-432-2324 — as early as possible. SAS sends the letter; I implement what it specifies;
- Illness or an emergency affecting your work needs a Dean's Excuse from your residential college Dean;
- Questions of any kind are welcome in office hours, on Ed, or by email.

Questions?